

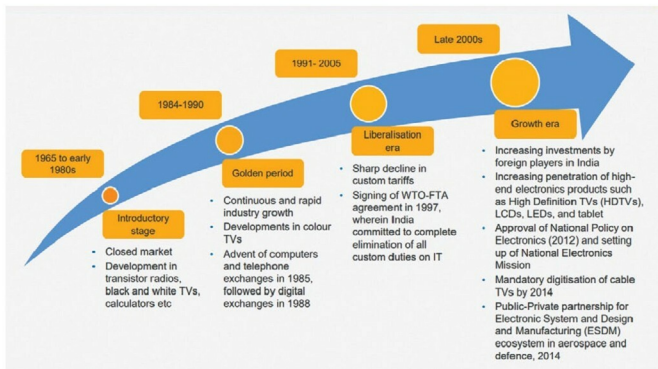


Fig. 1: Evolution of the Indian electronics industry (Source: Indian Electronics and Semiconductor Association report)

ing, AI and defense applications is likely to be the biggest growth driver in the electronics design market. In addition, there are a few emerging technologies that could emerge as key areas of focus in the near future. These include:

1. Next-generation wireless
2. Artificial intelligence
3. Silicon photonics

India has become the hub for electronics design with nearly 2000 chips being designed every year and around 500,000 engineers working in various aspects of electronic design starting from chip design and verification to different stages of product design. Though still lagging behind chief competitors in the Far East (China and Taiwan), the country has significant potential as it seeks to continually raise the bar and increase the breadth

and depth of its offering. At least 60 per cent of surveyed respondents (Fig. 3) feel that India has the potential to become a design house for the global electronics industry.

Greater digitisation across segments has led to the increased penetration of Indian electronic design houses in such application areas as LED illumination, communications and broadband equipment, and automotive electronics. The focus is on localising the technological changes while maintaining global competitiveness.

India's design strength is driven by:

1. Strong capabilities in embedded system design
2. Large number of electronics engineering graduates (even though the number of employable talent is

still around 10 per cent)

3. Increased demand due to a sizeable number of Indian SMEs and startups focusing on the IoT

According to survey participants, design services in demand (Fig. 4) are:

1. Embedded software design
2. End-to-end product design services
3. Electronic design automation (EDA)
4. Very-large-scale-integration (VLSI) design
5. Product certification and compliance

Market challenges

In recent years, independent design houses like Wipro and HCL have grown in stature and are increasingly involved in design and development for global firms. While conditions are

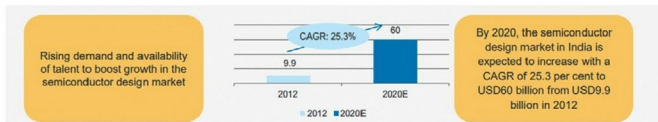


Fig. 2: Expected market growth of semiconductor design market in India (Source: Ministry of Electronics & IT, IESA, KPMG, Aranca Research)